



Siddaganga Institute of Technology, Tumakuru-572 103

(An Autonomous Institution affiliated to VTU, Belagavi, Approved by AICTE, New Delhi)

Even Semester Bachelor of Engineering Examinations July-Aug. 2024

Image Processing and Computer Vision

Time: 3 Hours

Max. Marks: 100

Note : 1. Revealing of Identity in any form in the answer book will be treated as malpractice.
2. Answer any five questions choosing one full question from each unit.

Unit - I

M BL CO PO PSO

- 1 a) With a neat block diagram list the fundamental steps in Digital Image Processing. Also develop a modified block diagram for a system to mark attendance of students entering into a class room using face recognition system. Identify suitable steps and explain.
- b) Check whether the pixels p and q in the image given in Fig. 1(b) are 4 or 8 or m-connected. Assume $V = \{70 - 80\}$. If connected, shown the shortest path and compute the path length.

p	70	78	80	82	78	77	78	82	78	
	80	76	82	83	82	80	82	82	78	
	80	82	80	73	80	80	82	82	80	
	80	80	83	83	80	82	83	82	82	
	87	80	82	82	82	80	82	80	78	
	89	80	80	80	80	80	72	73	80	q

Also find Euclidean, D_4 and D_8 distance between p and q.

OR

- 2 a) Develop connected component algorithm with 4-connectivity assuming $v=1$. Also apply the algorithm on the image shown in Fig. 2(a).

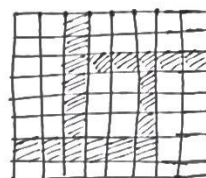


Fig. 2(a)

- b) Upsample the following 2×2 image using i) Nearest neighborhood interpolation
ii) Bilinear interpolation
- $$f(x, y) = \begin{bmatrix} 10 & 4 \\ 5 & 18 \end{bmatrix}$$
- c) A common measure of transmission for digital data is based rate. Each packet consists of a byte, start bit and a stop bit. How many minutes would it take to transmit a 1024×1024 image with 256 gray levels using i) 56k band modem
ii) 3000k band modem

Unit - II

- 3 a) With usual notations derive the filter mask of a Laplacian in spatial domain. Illustrate the limitations of Laplacian operator.
- b) Apply a 3×3 median filter for the image shown in Fig. 3(b). Comment on the results.

10	11	12	23	14	15
15	255	45	34	23	32
25	23	22	0	22	34
13	12	24	22	12	12

Fig. 3(b)

- c) Justify the need of image sharpening and mention applications of sharpening. Also, illustrate first and second derivatives used in sharpening.

OR

- 4 a) Analyze how first order derivatives are used for image sharpening. Also, bring out the Robert cross gradient, prewitt and sobel operators used for sharpening. 8 3 2 2 1
- b) Develop a basic global thresholding algorithm with automated threshold computation and list the limitations of this method. 6 3 2 2 1
- c) Analyze unsharp and highboost filtering with necessary expressions. 6 3 2 2 1

Unit - III

- 5 a) Analyze region feature descriptors with necessary diagrams and equations. 8 3 3 2 1
- b) Justify the statement “statistical moments of one variable are useful descriptors applicable to 1-D renditions of 2-D boundaries” with necessary equations. 6 3 3 2 1
- c) With necessary equations, analyze Fourier descriptors as boundary descriptors. 6 3 3 2 1

OR

- 6 a) Derive expression for Hotelling transform used to obtain feature descriptors. 8 3 3 2 1
- b) With necessary equations, analyze basic boundary descriptors used for image analysis. 6 3 3 2 1
- c) Analyze how 2D boundary is represented using 1D functional representation using signature. 6 3 3 2 1

Unit - IV

- 7 a) With necessary equations analyze intrinsic parameters of camera. 8 3 4 2 1
- b) Analyze how depth is computed using stereo camera with necessary equations. 6 3 4 2 1
- c) Derive the equations for perspective projection with pinhole camera and geometry diagram. 6 3 4 2 1

OR

- 8 a) With a neat block diagram, illustrate the working of 3D video system. 8 2 4 1 1
- b) Analyze the concepts of stereo vision and reconstruction with appropriate geometry diagram. 6 3 4 2 1
- c) Illustrate the concept of virtual view synthesis with necessary block diagram. 6 2 4 1 1

Unit - V

- 9 a) With a block diagram, illustrate the steps involved in designing a classification system. 8 2 5 1 1

- b) Perform hierarchical clustering using single linkage algorithm for the following features.

	x	y
1	4	4
2	8	4
3	15	8
4	24	4
5	24	12

- c) Compare the performance of a computer and that of a biological neural network in terms of speed, size, complexity, storage fault tolerance and control mechanism. 6 3 5 2 1

OR

- 10 a) Compare classification and regression. Analyze linear regression with an example and mention its limitation. 8 3 5 2 1

- b) Analyze the concept of k-means clustering and develop an algorithm for the same with necessary equations. 6 3 5 2 1

- c) Perform hierarchical clustering using average linkage algorithm for the following features.

	x	y
1	2	2
2	4	8
3	6	10
4	8	12
5	10	14

- 6 3 5 2 1